SUBJECT : PHYSICS

1. 1 mole O_2 gas and 1 mole N_2 gas are mixed in an enclosure of volume 5 litre and temperature 27° C. Find pressure of mixture :-

(1) 5 atm

(2) 15 atm

(3) 1 atm

(4) 10 atm

2. A diatomic gas expands in a process according to law $P^3 V^2 = \text{const}$. Then, find molar specific heat of gas in this process.

(1) $\frac{13}{2}R$

(2) $\frac{7}{2}R$

(3) $\frac{15}{2}R$

(4) $\frac{11}{2}R$

3. Two walls of thicknesses d_1 and d_2 and thermal conductivity k_1 and k_2 are in contact. In the steady state, if the temperatures at the outer surface are T_1 and T_2, the temperature at the common wall is

(1) $\frac{K_1T_1d_2 + K_2T_2d_1}{K_1d_2 + K_2d_1}$

(2) $\frac{K_1T_1 + K_2T_2}{d_1 + d_2}$

(3) $\left(\frac{K_1d_1 + K_2d_2}{T_1 + T_2}\right)$

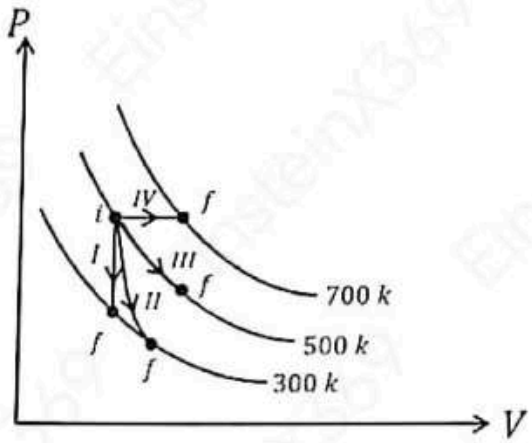
(4) $\frac{K_1d_1T_1 + K_2d_2T_2}{k_1d_1 + k_2d_2}$

4. Cooling rate of a sphere of 600 K at external environment (200K) is R. When the temperature of sphere is reduced to 400 K then cooling rate of the sphere becomes :

(1) $\frac{3}{16}R$ (2) $\frac{16}{3}R$

(3) $\frac{9}{27}R$ (4) None

5. Thermodynamic processes are indicated in the following diagram



Match the following:

Column-I	Column-II
P. Process I	a. Adiabatic
Q. Process II	b. Isobaric
R. Process III	c. Isochoric
S. Process IV	d. Isothermal

(1) P-d, Q-b, R-a, S-c

(2) P-a, Q-c, R-d, S-b

(3) P-c, Q-a, R-d, S-b

(4) P-c, Q-d, R-b, S-a

6. 50 gram of ice at 0°C is mixed with 50 gram of water at 60°C, final temperature of mixture will be :-

(1) 15°C

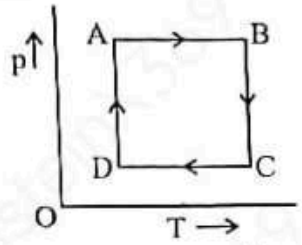
(2) 0°C

(3) 10°C

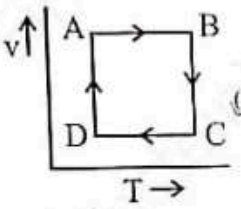
(4) 40°C

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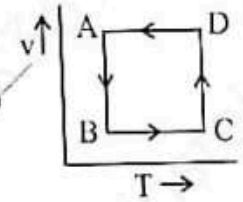
7. A cyclic process is shown on P-T diagram. Which of the curve shows the same process on a V-T diagram :-



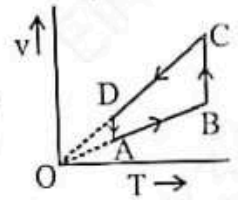
(1)



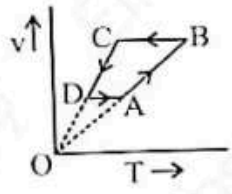
(2)



(3)



(4)



8. The ratio of specific heats (γ) of an ideal gas is given :-

(1) $\frac{1}{1-\left(\frac{R}{C_p}\right)}$

(2) $1+\frac{R}{C_v}$

(3) $\frac{C_p}{(C_p-R)}$

(4) All of these

9. Two star emit maximum radiation of wavelength 3600 Å and 4800 Å respectively then find ratio of their temperature is –

(1) 1 : 2 (2) 3 : 4

(3) 4 : 3 (4) 2 : 1

10. On increasing the temperature of a gas filled in a closed container by 1°C its pressure increases by 0.4%, then initial temperature of the gas is -

(1) 25°C (2) 250°C

(3) 250 K (4) 2500°C

11. A body takes 4 minutes to cool from 100°C to 70°C. To cool from 70°C to 40°C it will take- (room temperature is 15°C)

(1) 7 minutes

(2) 6 minutes

(3) 5 minutes

(4) 4 minutes

12. At what temperature will the oxygen molecules have the same root mean square speed as hydrogen molecules at 200 K-

(1) 800 K

(2) 1600 K

(3) 2400 K

(4) 3200 K

13. The temperature gradient in a rod of 0.5 m long is 80°C/m. If the temperature of hotter end of the rod is 30°C, then the temperature of the cooler end is :-

(1) 40°C (2) –10°C

(3) 10°C (4) 20°C

14. Heat is transmitted from higher to lower temperature through actual mass motion of the molecules in:-

(1) Conduction

(2) Convection

(3) Radiation

(4) All of above

15. A black body is at a temperature of 2880 K. The energy of radiation emitted by the body at wavelength 500 nm is U_1, at wavelength 1000 nm is U_2 and that at 2000 nm is U_3. Wien's constant, $b = 2.88 \times 10^6$ nmK. Which of the following is correct ?

(1) $U_1 = 0$ (2) $U_3 = 0$

(3) $U_1 > U_2$ (4) $U_2 > U_1$

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16. The molar heat capacity for an ideal gas -

(a) is zero for adiabatic process.

(b) is infinite for an isothermal process

(c) depends on the nature of gas for a process in which either volume or pressure is constant

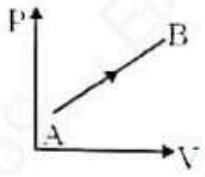
(d) is equal to the product of the molecular weight and gram specific heat capacity for any process.

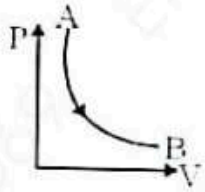
Choose the correct statements :-

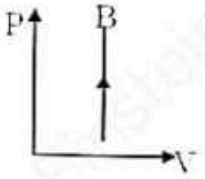
(1) a, b (2) c,d

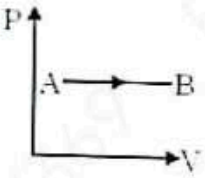
(3) a, b, c & d (4) b, c, d

17. The process $\Delta U = 0$, for an ideal gas can be best represented in the form of a graph :-

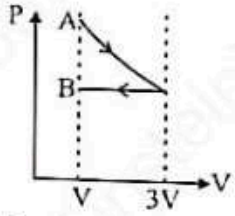
(1) 

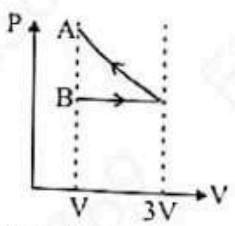
(2) 

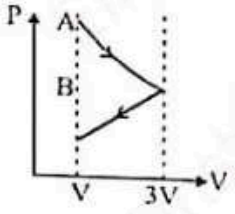
(3) 

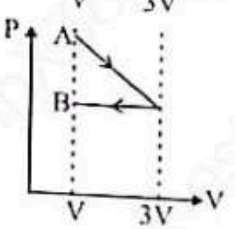
(4) 

18. One mole of an ideal gas goes from an initial state A to final state B via two processes: It first undergoes isothermal expansion from volume V to 3V and then its volume is reduced from 3V to V at constant pressure. The correct P-V diagram representing the two processes is :-

(1) 

(2) 

(3) 

(4) 

19. If water is heated on top of Mount Everest, where atmospheric pressure is very low, water will start boiling at temp : -

(1) At temperature less than 100°C

(2) At fixed 100°C

(3) At temperature more than 100°C

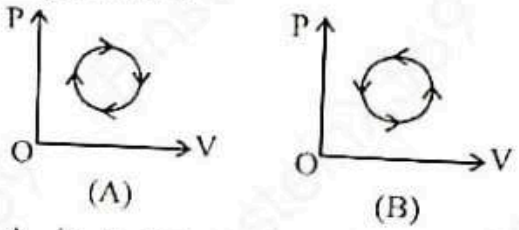
(4) Water will freeze on heating

20. An ideal heat engine has an efficiency η. The coefficient of performance of the engine when driven backward will be

(1) $1-(1/\eta)$ (2) $\eta/(1-\eta)$

(3) $(1/\eta)-1$ (4) $1/(1-\eta)$

21. If the P-V diagrams of two thermodynamics devices working in a cyclic process are as shown in the figure, then :-



(1) A is a heat engine, B is a heat pump/refrigerator

(2) B is a heat erfgine, A is a heat pump/refrigerator

(3) Both A and B are heat engines

(4) Both A and B are heat pumps/refrigerator

22. Which of the following is correct according to Claussius statement of second law of thermodynamics.

(1) It is possible to transfer heat from a low temp. reservoir to a high temp. reservoir.

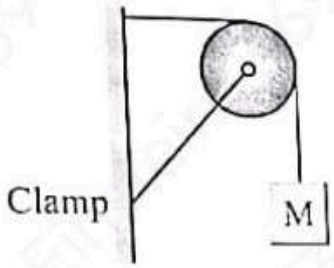
(2) No heat engine can have 100% efficiency.

(3) Moving heat from a cold region to a warm region requires an external energy or work input.

(4) None of these

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A string of negligible mass going over a clamped pulley of mass m supports a block of mass M as shown in the figure. The force on the pulley by the clamp is given by :



(1) $\sqrt{2}$ mg

(2) $\sqrt{2}$ Mg

(3) $\sqrt{(M+m)^2+m^2g}$

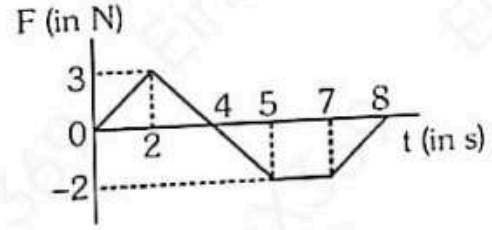
(4) $g\sqrt{(M+m)^2+M^2}$

24. At what temperature, rms velocity of O_2 molecules will be 1/3 that of H_2 molecules at -3 °C?

(1) 90 K (2) 1167°C

(3) -3°C (4) 207°C

25. Force-time graph for the motion of a body is shown in fig. Change in linear momentum between 0 to 8 s is :-



(1) Zero (2) 4 N-s

(3) 8 N-s (4) 6 N-s

26. A rocket of mass 60 kg. is fired in a gravity free space is ejecting gases with velocity 300 m/s at the rate of 2 kg/s. What will be the initial acceleration of the rocket ?

(1) 30 m/s^2 (2) 5 m/s^2

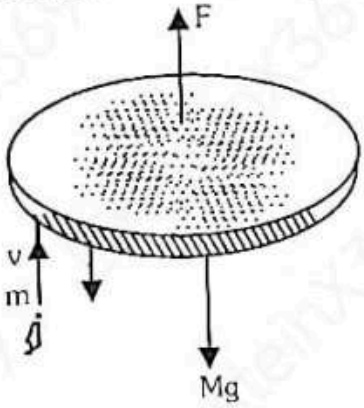
(3) 10 m/s^2 (4) 15 m/s^2

27. A particle moves in x-y plane under the influence of a force such that its linear momentum $\vec{P}(t) = A\left[\hat{i}\cos(kt) - \hat{j}\sin(kt)\right]$ where A and k are constants. The angle between the force and the momentum is :-

(1) 0° (2) 30°

(3) 45° (4) 90°

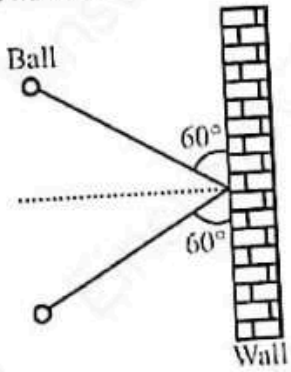
28. A disc of mass 1·0 kg kept floating horizontally in air by firing bullets of mass 0·05 kg each vertically at it, at the rate of 10 per second. If the bullets rebound with the same speed, the speed with which these are fired will be– ($g = 9.8$ m/s^2)



(1) 0·098 m/s (2) 0·98 m/s

(3) 9·8 m/s (4) 98·0 m/s

29. A ball of mass 3 kg, moving with a speed of 100 m/s, strikes a wall at an angle 60° (as shown in figure). The ball rebounds at the same speed and remains in contact with the wall for 0.2 sec; the force exerted by the ball on the wall is :-



(1) $1500\sqrt{3}$ N (2) 1500 N

(3) $300\sqrt{3}$ N (4) 300 N

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30. The force exerted by the floor of an elevator on the foot for a person standing there, is more than the weight of the person, if the elevator is :
A. Going up with increasing speed
B. Going down with decreasing speed

(1) A is true but B is false

(2) A is false but B is true

(3) Both A and B are true

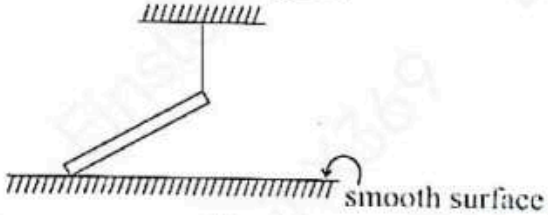
(4) Both A and B are false

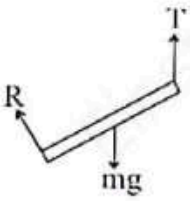
31. A rocket of initial mass 1500 kg. ejects gas at a constant rate of 10kg/s. with a relative speed of 5 km/s. What is the acceleration of the rocket 50 seconds after the blast, neglecting gravity :-

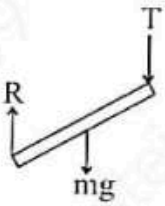
(1) $10\ ms^{-2}$ (2) $25\ ms^{-2}$

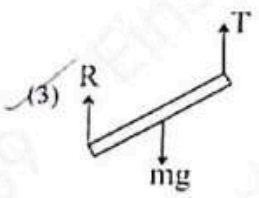
(3) $50\ ms^{-2}$ (4) $100\ ms^{-2}$

32. Which figure represents the correct F.B.D. of rod of mass m as shown in figure :



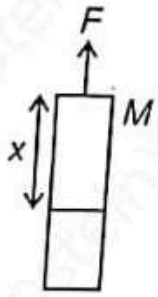
(1) 

(2) 

(3) 

(4) None of these

33. A uniform bar of mass M and length ℓ is carried upward by applying force F on its upper end. The force at the distance x from its upper end is



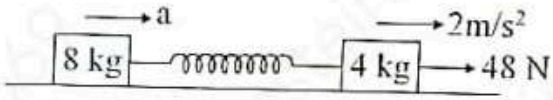
(1) $\frac{F(\ell - x)}{\ell}$

(2) $\frac{Fx}{\ell}$

(3) $\frac{(F - Mg)x}{\ell}$

(4) $\frac{(F - Mg)(\ell - x)}{\ell}$

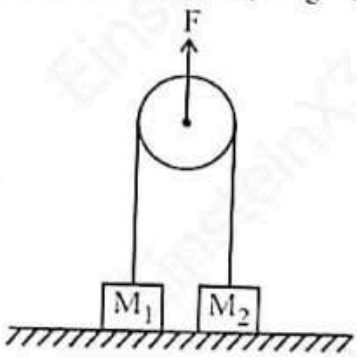
34. Find acceleration of 8 kg block when acceleration of 4 kg block is $2\ m/s^2$ in given situation –



(1) $2\ m/s^2$ (2) $8\ m/s^2$

(3) $5\ m/s^2$ (4) $4\ m/s^2$

35. In shown figure $M_1 = 4kg$ and $M_2 = 8kg$ and the pulley is moveable. At $t = 0$, both masses touch the ground and the string is taut. A vertically upward force $F = 2t$ is applied to the pulley. The time when M_2 is lifted off the ground is :- (take $g = 10\ m/s^2$)

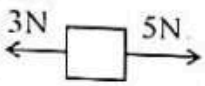


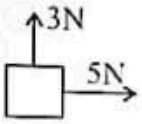
(1) 40 sec (2) 80 sec

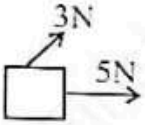
(3) 120 sec (4) 160 sec

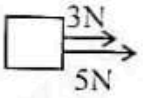
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36. Two forces, one with a magnitude of 3N and the other with a magnitude of 5N, are applied to an object. For which orientations of the forces shown in the diagrams is the magnitude of the acceleration of the object the least ?

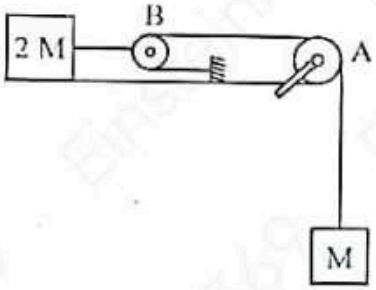
(1) 

(2) 

(3) 

(4) 

37. Consider the situation as shown in the figure:-



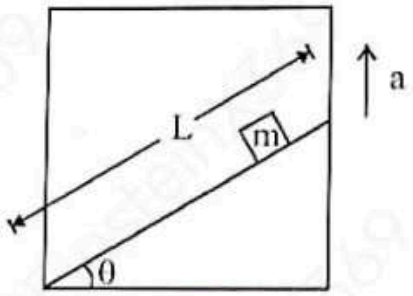
(1) The acceleration of M is 2g/3

(2) The tension in the string connecting A and B is Mg/3

(3) The reaction on pulley A is $\sqrt{2}$ Mg/3 at an angle 45° with the horizontal

(4) All the options are correct

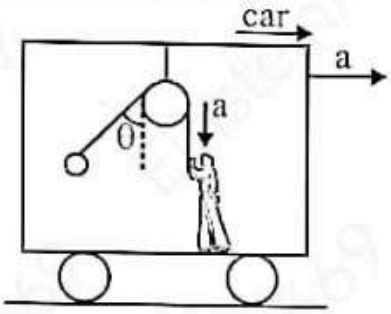
38. A block is sliding along an inclined plane as shown in figure. If the acceleration of chamber is a as shown in the figure. The time required to cover a distance L along incline is :-



(1) $\sqrt{\frac{2L}{g\sin\theta - a\cos\theta}}$ (2) $\sqrt{\frac{2L}{g\sin\theta + a\sin\theta}}$

(3) $\sqrt{\frac{2L}{g\sin\theta + a\cos\theta}}$ (4) $\sqrt{\frac{2L}{g\sin\theta}}$

39. A bob is hanging over a pulley inside a car through a string. The second end of the string is in the hand of a person standing in the car. The car is moving with constant acceleration 'a' directed horizontally as shown in figure. Other end of the string is pulled with constant acceleration 'a' (relative to car) vertically. The tension in the string is equal to :



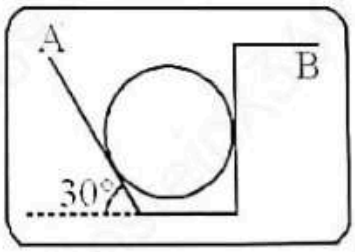
(1) $m\sqrt{g^2+a^2}$

(2) $m\sqrt{g^2+a^2} - ma$

(3) $m\sqrt{g^2+a^2} + ma$

(4) $m(g+a)$

40. The 50kg homogeneous smooth sphere rests on the 30° incline A and bears against the smooth vertical wall B. Calculate the contact forces at A and B :



(1) $N_B = \frac{1000}{\sqrt{3}}N, N_A = \frac{500}{\sqrt{3}}N$

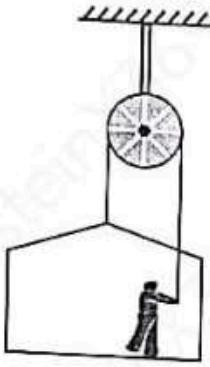
(2) $N_A = \frac{1000}{\sqrt{3}}N, N_B = \frac{500}{\sqrt{3}}N$

(3) $N_A = \frac{100}{\sqrt{3}}N, N_B = \frac{500}{\sqrt{3}}N$

(4) $N_A = \frac{1000}{\sqrt{3}}N, N_B = \frac{50}{\sqrt{3}}N$

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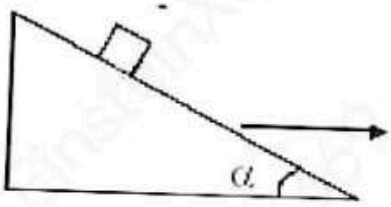
41. A man is raising himself and the crate on which he stands with an acceleration of 5 ms^{-2} by a massless rope–and–pulley arrangement. Mass of the man is 100 kg and that of the crate is 50 kg. If g = 10 ms^{-2}, the tension in the rope is



(1) 2250 N (2) 1125 N

(3) 750 N (4) 375 N

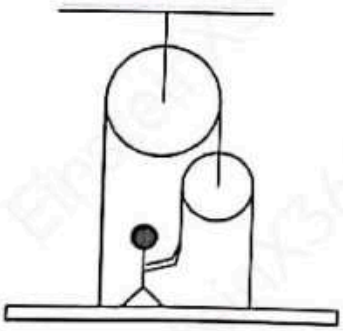
42. A block is kept on a frictionless inclined surface with angle of inclination 'α'. The incline is given an acceleration 'a' to keep the block stationary. Then a is equal to



(1) g (2) g tan α

(3) g / tan α (4) g cosec α

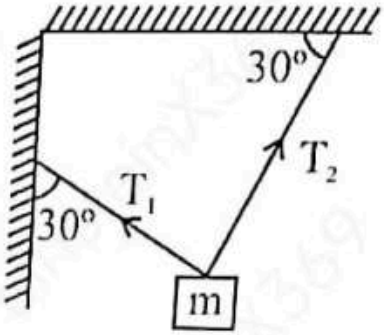
43. In the given diagram, with what force must the man pull the rope to hold the plank in position? Mass of the man is 80 kg. Neglect the weights of plank, rope and pulley. Take g = 10 ms^{-2}



(1) 200 N (2) 300 N

(3) 600 N (4) 150 N

44. Calculate T_1 & T_2 :-



(1) $\frac{\sqrt{3}}{2}mg, \frac{mg}{2}$

(2) $mg, \frac{\sqrt{3}}{2}mg$

(3) $\frac{\sqrt{3}}{2}mg, \frac{\sqrt{3}}{2}mg$

(4) $\frac{mg}{2}, \frac{mg}{2}$

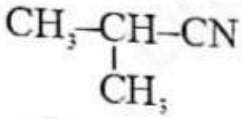
45. A vessel has 6g of oxygen at pressure P and temperature 400K. A small hole is made in it so that oxygen leaks out. How much oxygen leaks out if the final pressure is P/2 and temperature is 300 K?

(1) 3g (2) 2g

(3) 4g (4) 5g

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46. Comman name of the given compound is :-



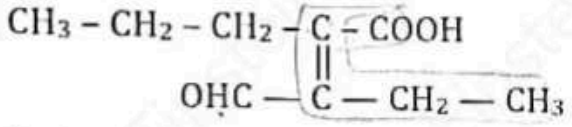
(1) 2-Methylpropanenitrile

(2) Isobutylcyanide

(3) 2-Cyanopropane

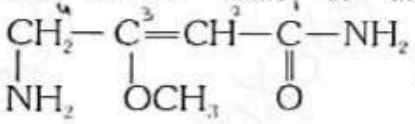
(4) Isobutyronitrile

47. The number of carbon atoms in the principle carbon chain of the given compound is :-



(1) 7 (2) 5 (3) 4 (4) 6

48. The IUPAC name of the compound



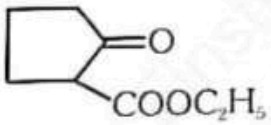
is :-

(1) 4–Amido-2–methoxy–1–amino–2–butene

(2) 4–Amino–3–methoxy–2–butXenamide

(3) 2–Methoxy–1,4–diamino–2–butenal

(4) 1–Amino–2–methoxy–3–amido propene

49.  has the IUPAC name :

(1) Ethyl–2–oxo cyclopentane carboxylate

(2) 2–Cyclopentanone–1–carbethoxy

(3) 2–Ethylcarbonate cyclopentanone

(4) 1–Keto–2–carbethoxy cyclopentanone

50. In $\overset{1}{CH_2} = \overset{2}{C} = \overset{3}{CH} - \overset{4}{CH_3}$ molecule, the hybridization of carbon 1, 2, 3 and 4 respectively are :

(1) sp^3, sp, sp^3, sp^3 (2) sp^2, sp^2, sp^2, sp^3

(3) sp^2, sp, sp^2, sp^3 (4) sp^2, sp^3, sp^2, sp^3

51. Which of the following statement is **wrong** for homologues :-

(1) They have same chemical properties

(2) Two successive members differ by a $(-CH_2)$ group.

(3) They have different functional group.

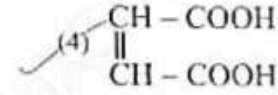
(4) They have same general formula.

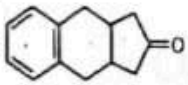
52. Which of the following is crotonic acid :

(1) $CH_2 = CH-COOH$

(2) $C_6H_5-CH=CH-COOH$

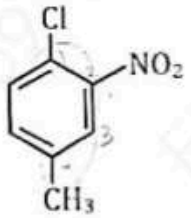
(3) $CH_3-CH=CH-COOH$

(4) 

53. Degree of unsaturation in  is :

(1) 3 (2) 4 (3) 7 (4) 6

54. The IUPAC name for

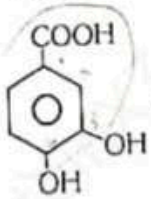


(1) 1-Chloro-2-nitro-4-methylbenzene

(2) 1-Chloro-4-methyl-2-nitrobenzene

(3) 2-Chloro-1-nitro-5-methylbenzene

(4) m-Nitro-p-chlorotoluene

55. The IUPAC name of  is :-

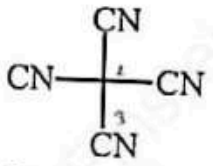
(1) 3, 4–Dihydroxy benzoic acid

(2) 2, 3–Dihydroxy benzoic acid

(3) 4–Carboxy–2–hydroxy phenol

(4) 4–Carboxy benzene–1, 2–diol

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56. 

Correct IUPAC name :-

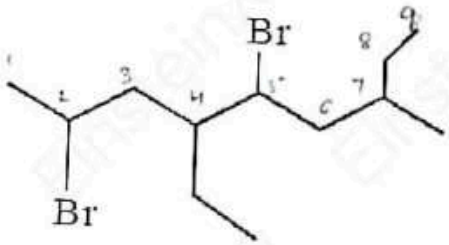
(1) 2, 2-Dicyano- 1, 3-Propandinitrile

(2) 1, 3-Dicyano propen-1, 3-Dinitrile

(3) Methan tetra carbonitrile

(4) All of the above

57. The systematic name of given structure is



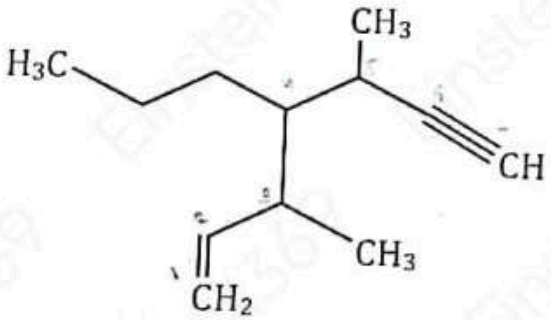
(1) 2, 5-dibromo-4, 7-diethyl octane

(2) 4, 7-dibromo-2, 5-diethyl octane

(3) 5, 8-dibromo-6-ethyl-3methyl nonane

(4) 2, 5-dibromo-4-ethyl-7-methyl nonane

58. The IUPAC name of the following compound is :



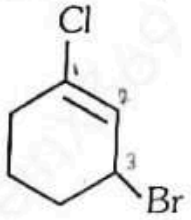
(1) 3,5-Dimethyl-4-propylhept-6-en-1-yne

(2) 3-Methyl-4-(3-methylprop-1-enyl)-1-heptyne

(3) 3-Methyl-4-(1-methylprop-2-ynyl)-1-heptene

(4) 3,5-Dimethyl-4-propylhept-1-en-6-yne

59. The IUPAC name of the compound is



(1) 6-bromo-2-chlorocyclohexene

(2) 3-bromo-1-chlorocyclohexene

(3) 1-bromo-3-chlorocyclohexene

(4) 2-bromo-6-chlorocyclohex-1-ene

60. The IUPAC name of urea is :-

(1) Diamino ketone

(2) Amino methanamide

(3) Amino ethanamide

(4) Amino aceta mide

61. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Dipole moment of NF_3 is less than that of NH_3

Reason (R): Polarity of N—F bond is less than that of N—H bond.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

62. **Assertion :** Geometry of ICl_3 is tetrahedral.

Reason : Its shape is T-shape, due to the presence of two lone pairs.

(1) If assertion is true but reason is false

(2) If assertion is false but reason is true

(3) If both assertion and reason are true and the reason is the correct explanation of assertion

(4) If both assertion and reason are true and the reason is not the correct explanation of assertion

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63. Match the species in column-I with the bond order in column-II

	Column-I		Column-II
(i)	NO	(a)	1.5
(ii)	CO	(b)	2.0
(iii)	O_2^-	(c)	2.5
(iv)	O_2	(d)	3.0

(1) i-a, ii-b, iii-d, iv-c

(2) i-c, ii-d, iii-a, iv-b

(3) i-c, ii-d, iii-b, iv-a

(4) i-d, ii-c, iii-a, iv-b

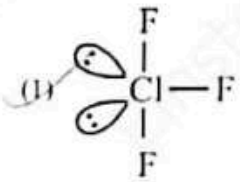
64. Match the compounds given in column I with the hybridisation and shape given in column II and mark the **correct** option.

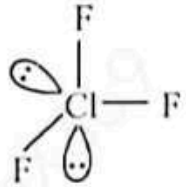
Column - I		Column - II	
(a)	XeF_6	(i)	Distorted octahedral
(b)	XeO_3	(ii)	Square planar
(c)	$XeOF_4$	(iii)	Pyramidal
(d)	XeF_4	(iv)	Square pyramidal

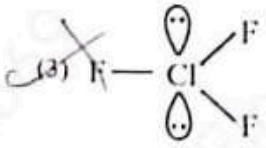
	(a)	(b)	(c)	(d)
(1)	(i)	(iii)	(iv)	(ii)
(2)	(i)	(ii)	(iv)	(iii)
(3)	(iv)	(iii)	(i)	(ii)
(4)	(iv)	(i)	(ii)	(iii)

(1) 1 (2) 2 (3) 3 (4) 4

65. The correct shape of ClF_3 is-

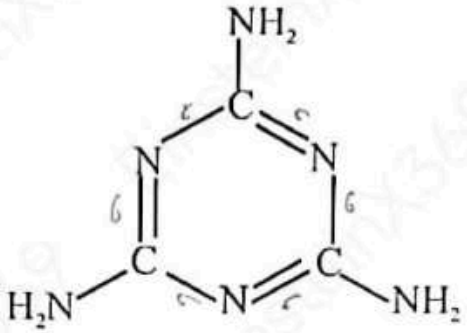
(1) 

(2) 

(3) 

(4) None of these

66. Find the number of sigma bond in melamine (given as) :



(1) 9

(2) 12

(3) 15

(4) 18

67. Consider the following statements:

I. A sigma (σ) bond is formed when two s-orbitals overlap

II. A pi(π) bond is formed when two p-orbitals axially overlap

III. A σ-bond is weaker than π-bond

Which of the above statement is/are correct ?

(1) I and II

(2) II and III

(3) I alone

(4) II alone

68. Predict the correct order of repulsion among the following.

(1) Lone pair - bond pair > bond pair - bond pair > lone pair - lone pair.

(2) Lone pair - lone pair > lone pair - bond pair > bond pair - bond pair.

(3) Lone pair - lone pair > bond pair - bond pair > lone pair - bond pair.

(4) Bond pair - bond pair > lone pair - bond pair > lone pair - lone pair.

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69. Consider the molecules CH_4, NH_3 and H_2O. Which of the given statements is false ?

(1) The H – C – H bond angle in CH_4 is larger than the H – N – H bond angle in NH_3.

(2) The H – C – H bond angle in CH_4, the H – N – H bond angle in NH_3, and the H – O – H bond angle in H_2O are all greater than 90°.

(3) The H – O – H bond angle in H_2O is larger than the H – C – H bond angle in CH_4.

(4) The H – O – H bond angle in H_2O is smaller than the H – N – H bond angle in NH_3.

70. The correct increasing order of bond angle is

(1) CO_2, SO_2, BF_3, CH_4

(2) CH_4, SO_2, BF_3, CO_2

(3) BF_3, CH_4, CO_2, SO_2

(4) CO_2, CH_4, BF_3, SO_2

71. Which of the following is true ?

(1) Bond order $\propto \frac{1}{\text{Bond length}} \propto$ Bond energy

(2) Bond order $\propto$ Bond length $\propto \frac{1}{\text{Bond energy}}$

(3) Bond order $\propto \frac{1}{\text{Bond length}} \propto \frac{1}{\text{Bond energy}}$

(4) Bond order $\propto$ Bond length $\propto$ Bond energy

72. Wrong statement is :

(1) sp^3d^2 and d^2sp^3 hybridisation only differ in way of principle quantum number

(2) HNO_3 has covalent and coordinate bond both

(3) I_3^- and CO_2 have same shape

(4) $CsBr_3$ contain Cs^+, Br^- ions

73. Match column-I with column-II and select the correct answer :-

	Column-I (Molecules)		Column-II (Property)
(P)	IF_7	(1)	Planar and polar
(Q)	SO_2	(2)	Nonplanar and polar
(R)	SF_4	(3)	Planar and nonpolar
(S)	CO_2	(4)	Nonplanar and nonpolar

	(P)	(Q)	(R)	(S)
(1)	4	1	2	3
(2)	4	2	1	3
(3)	3	1	2	4
(4)	3	1	4	2

74. **Assertion :-** In SO_3 molecule bond dissociation energy of all S=O bonds are not equivalent.

Reason :- SO_3 molecule is having two types of $2p\pi-3p\pi$ & $2p\pi-3d\pi$, π-bonds.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

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75. Given below are two statements:

Statement-I : Lewis theory explains the shape of a molecules.

Statement-II : It explain the relative stability and energy of the molecules.

In the light of the above statements, choose the most appropriate answer from the options given below:

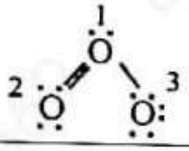
(1) Statement-I and II both are correct.

(2) Both Statement-I and II are incorrect.

(3) Statement-I is incorrect and II is correct.

(4) Statement-I is correct while II is incorrect.

76. Lewis structure of O_3 is shown



Column-I	Column-II (Formal charges)
(a) The central O atom marked 1	(i) −1
(b) The end O atom marked 2	(ii) 0
(c) The end O atom marked 3	(iii) +1

(1) a–i; b–ii; c–iii (2) a–iii; b–ii; c–i

(3) a–ii; b–i; c–iii (4) a–i; b–iii; c–ii

77. Which condition favours the bond formation:

(1) Maximum attraction and maximum potential energy

(2) Minimum attraction and minimum potential energy

(3) Minimum potential energy and maximum attraction

(4) None of the above

78. Dry ice is.

(1) solid SO_2 (2) solid NH_3

(3) solid O_2 (4) solid CO_2

79. London dispersion force present between which pair.

(1) HBr & HBr (2) HCl & HCl

(3) Cl_2 & HCl (4) Br_2 & Br_2

80. Which of the following is the correct order of stability ?

(1) $H_2 > H_2^+ > He_2 > He_2^+$

(2) $H_2 > He_2^+ > H_2^+ > He_2$

(3) $H_2 > H_2^+ > He_2^+ > He_2$

(4) $H_2 > He_2 > He_2^+ > He_2^+$

81. Give the correct order of strength of H-bond

(a) O – H —— O (b) F – H ------ F

(c) Cl – H ----- Cl (d) N – H ----- N

(1) a > b > c > d

(2) b > a > c > d

(3) b > a > d > c

(4) b > a > c = d

82. Ionic conductances of hydrated M^+ ions are in the order:-

(1) $Li^+(aq) > Na^+(aq) > K^+(aq) > Rb^+(aq) > Cs^+(aq)$

(2) $Li^+(aq) > Na^+(aq) < K^+(aq) < Rb^+(aq) < Cs^+(aq)$

(3) $Li^+(aq) > Na^+(aq) > K^+(aq) > Rb^+(aq) < Cs^+(aq)$

(4) $Li^+(aq) < Na^+(aq) < K^+(aq) < Rb^+(aq) < Cs^+(aq)$

83. London Dispersion force is mainly present between :-

(1) Polar + Polar molecules

(2) Polar + Nonpolar molecules

(3) Nonpolar + Non-polar molecules

(4) Ionic

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84. Which of the following is the incorrect order of bond strength ?

(1) 1s–1s < 2p–2p

(2) 2s–2s < 2p–2p

(3) 1s–2s < 1s–2p

(4) 3p–3p > 4p–4p

85. Octet rule is not valid for the molecule :

(1) CO_2 (2) NO

(3) O_2 (4) PCl_3

86. Consider the following molecules ;

H_2O H_2S H_2Se H_2Te

I II III IV

Arrange these molecules in increasing order of bond angles.

(1) I < II < III < IV (2) IV < III < II < I

(3) I < II < IV < III (4) II < IV < III < I

87. The dipole moment of o, p and m-dichlorobenzene will be in the order:

(1) o > p > m (2) p > o > m

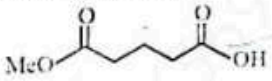
(3) m > o > p (4) o > m > p

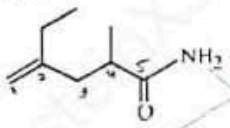
88. Which molecule having covalent bond?

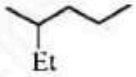
(1) KI_3 (2) NaI

(3) KF (4) NaBr

89. Match the following :-

(a)  (p) 4C in p.c.c.

(b)  (q) 6C in p.c.c.

(c)  (r) 5C in p.c.c.

(1) (a – p, b – q, c – r)

(2) (a – r, b – q, c – p)

(3) (a – p, b – r, c – q)

(4) (a – q, b – p, c – r)

90. Which of following has/have general formula $C_nH_{2n}O$:-

(i) Alkanal

(ii) Alkoxyalkene

(iii) Cycloalkanone

(iv) Alkenol

(1) i, ii, iii, iv (2) i, iii

(3) i, ii, iv (4) ii, iv

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91. In two kingdom classification. The main character of the members of plantae was :-

(1) Autotrophic in nature

(2) Presence of cell wall

(3) Starch as reserve food

(4) Non motile organisms

92. What is the role of mesosomes in bacteria?

(1) Cell wall synthesis

(2) Respiration

(3) Helps in DNA replication

(4) All of these

93. Read the given statements

Statement A : Aristotle classified plants on the basis of habitat of the plants.

Statement B : Organisms with loose tissue body organisation are placed in Kingdom fungi.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements are incorrect

(4) Both statements are correct

94. In R.H. Whittaker's classification *Amoeba* and *Paramoecium* are placed in the kingdom_____

(1) Bacteria

(2) Protista

(3) Fungi

(4) Monera

95. Which of the following kingdoms contain organisms with non cellulosic cell wall?

(1) Monera

(2) Fungi

(3) Protista

(4) More than one option is correct

96. Select the odd one w.r.t. criteria considered by R.H. Whittaker to classify living organisms.

(1) Body organisation

(2) Phylogeny

(3) Locomotion

(4) Mode of nutrition

97. All protozoans are -

(1) Saprophytes only

(2) Parasites only

(3) Predators only

(4) Heterotrophs (parasite/predators) only

98. Bacteria participate in -

(1) Making curd from milk

(2) N_2 - fixation in legume roots

(3) Disease causing in plants, animals and human beings

(4) All of these

99. Euglenoids eg. Euglena are found -

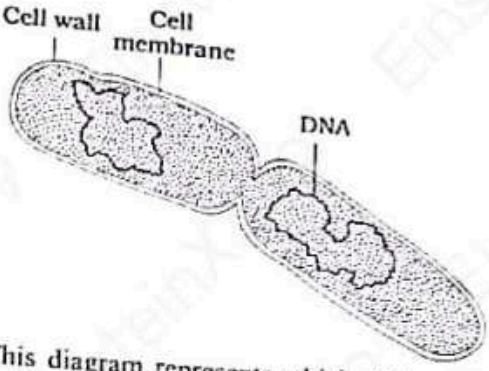
(1) In fresh running water

(2) In fresh stagnant water

(3) In marine environment

(4) In both fresh and marine water

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100. 

This diagram represents which of the following organism?

(1) Slime mould

(2) *Euglena*

(3) Dividing fungi

(4) Dividing bacterium

101. Which of the following character is similar in cyanobacteria and green plants :-

(1) Nitrogen fixation

(2) 80 s ribosme

(3) Chlorophyll 'a'

(4) Composition of cell wall

102. Mycoplasma differs from bacteria in the absence of :-

(1) Ribosome

(2) Cell membrane

(3) DNA

(4) Cell wall

103. Read statements A to D :

(A) Golden Algae are classified under chrysophytes.

(B) Gonyaulax & Diatoms are classified under Dinoflagellates

(C) In Diatoms the cell wall form thin overlapping shell, which fit together as in soap box.

(D) Dinoflagellates have two flagella.

How many statement are wrong ?

(1) one (2) two

(3) three (4) four

104. Toxins released by which group of protista is the cause of death of other marine animals such as fishes ?

(1) Chrysophytes

(2) Euglenoids

(3) Desmids

(4) Dinoflagellates

105. **Assertion:** Spores are extremely resistant and Survive for many years.

Reason: The spores prossess true walls.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

106. In which of the following fungi karyogamy is followed by meiosis and produces exogenous spores.

(1) Neurospora

(2) Agaricus

(3) Alternaria

(4) Saccharomyces

107. Identify the correct match and select the correct option from options given below :

(A) Yeast : Unicellular fungi : Basidiomycetes

(B) *Albugo* : Parasitic fungi : Phycomycetes

(C) *Puccinia* : Rust fungi : Basidiomycetes

(D) Truffles : Edible mushroom : Ascomycetes

(1) A, B, C and D

(2) B and C

(3) B, C and D

(4) A, B and C

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108. Match the column I with II :-

Column–I		Column–II	
(i)	Ascomycetes	(a)	*Alternaria*
(ii)	Basidiomycetes	(b)	*Mucor*
(iii)	Phycomycetes	(c)	*Ustilago*
(iv)	Deuteromycetes	(d)	*Penicillium*

(1) i - a, ii - b, iii - c, iv - d

(2) i - b, ii - c, iii - a, iv - d

(3) i - d, ii - c, iii - b, iv - a

(4) i - d, ii - c, iii - a, iv - b

109. Which of the following is correct for class that includes Bracket fungi and puff balls.

(A) Asexual spores are common

(B) Sexual organs are present

(C) Exogenous basidiospores

(D) Somatogamy between same genotypes

(E) Karyogamy and meiosis in basidium

(1) A, C & E (2) C, D & E

(3) A, B, C & D (4) C & E only

110. Consider the following four statements (A-D) and select the option which includes all the correct options only:-

(A) M.W Beijerineck demonstrated that the extract of the infected plant of Tobacco could cause infection.

(B) W.M Stanley showed that virus are crystals consist largely of proteins.

(C) Lichens are pioneer vegetation on bare rock.

(D) Virus was discovered by Ivanowsky.

(1) A, B and D (2) B, C and D

(3) C and D (4) A, B, C and D

111. The infectious abnormally folded proteins, similar in size to viruses are known as-

(1) Viroids (2) Prions

(3) Virions (4) Rickettsia

112. Mark the statements as True(T) or False(F) respectively

A. Viruses are inert outside their specific host

B. Virus contain only DNA

C. The protein coat of viruses is made of subunits called prion

Identify the correct option :

(1) TTT (2) T,F,T

(3) F,T,T (4) T,F,F

113. How many diseases or symptoms are caused by virus from the given box?

Stunted growth	Mosaic disease
Leaf rolling	Canker disease
Cholera	Tetanus
Typhoid	

(1) Three (2) Four

(3) Five (4) Two

114. Select the incorrect statement about viroids :-

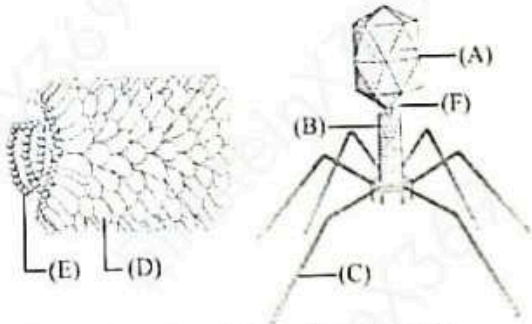
(1) Viroids are free RNA infectious agents

(2) It cause potato spindle tuber disease

(3) It is smaller than virus

(4) The RNA of the viroids have high molecular weight

115.



Identify (A), (C) and (E) in these figures :-

(1) A = Sheath, C = Tail fibres, E = DNA

(2) A = RNA, C = Tail fibres, E = Collar

(3) A = Head, C = Tail fibres, E = RNA

(4) A = Capsid, C = Sheath, E = RNA

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116. TMV has :-

(1) ds RNA (2) ds DNA

(3) ss RNA (4) ss DNA

117. Most names in biological nomenclature of living organisms are taken from which language? (n)

(1) Hindi (2) Latin

(3) German (4) French

118. Select the correctly written scientific name of Mango which was first described by Carolus Linnaeus

(1) *Mangifera indica* Car. Linn.

(2) *Mangifera indica* Linn.

(3) *Mangifera indica*

(4) *None of these*

119. Which of the following category of taxonomic hierarchy will have maximum number of common characters ?

(1) Family (2) Order

(3) Division (4) Class

120. The housefly belongs to which family in taxonomical classification ?

(1) Musca (2) Diptera

(3) Muscidae (4) Insecta

121. Read the following statements.

(A) Each genus may have one or more than one specific epithet.

(B) Genera are aggregates of closely related species.

(C) Family has a group of related genera with still more number of similarities as compared to genus and species.

How many above statements are correct and incorrect ?

(1) 3 correct, 0 incorrect

(2) 2 correct, 1 incorrect

(3) 1 correct, 2 incorrect

(4) 0 correct, 3 incorrect

122. Choose the correct from followings :-

(1) *Mangifera*, is a name of species

(2) Polymoniales, is a name of family

(3) Dicotyledonae, is a name of division

(4) *Triticum*, is a name of genus

123. Division

↑

Class

↑

C

↑

B

↑

Genus

↑

A

In given taxonomic sequence A, B & C could be?

	A	B	C
(1)	Sub species	Family	Kingdom
(2)	Species	Order	Family
(3)	Species	Family	Order
(4)	Order	Family	Kingdom

(1) 1 (2) 2

(3) 3 (4) 4

124. Sapindales is the order of :

(1) Mango (2) Maize

(3) Wheat (4) Corn

125. What is the similarity betweeen panthera and solanum ?

(1) Both are group of plants

(2) Both are families

(3) Both are genus

(4) Both are order

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126. Classes comprising animals, like fishes, amphibians, reptiles, birds along with mammals, constitute the next higher category called

(1) Phylum (2) Genus

(3) Kingdom (4) Order

127. Order is the group of related families which has ____ characters in common in comparison to families of _____.

(1) Less, other orders

(2) More, other orders

(3) Less, other classes

(4) More, other classes

128. The basis of modern taxonomic studies are :-

(a) External and internal structure

(b) Structure of cell

(c) Development process

(d) Ecological informations

(1) a and b

(2) a, b and c

(3) a, c and d

(4) All a, b, c and d

129. What is the basic unit of classification?

(1) Family (2) Order

(3) Species (4) Genus

130. Plant families like convolvulaceae, solanaceae are included in which order :

(1) Poales

(2) Poaceae

(3) Polymoniales

(4) Sapindales

131. Study of different kinds of organisms and their diversities with relationship among them is called ?

(1) Taxonomy (2) Systematics

(3) Classification (4) Identification

132. Which of the following organisms does not reproduce ?

(1) Mule

(2) Worker bee

(3) Infertile human female

(4) All of these

133. "The Darwin of 20th century is :

(1) Huxley (2) Hutchinson

(3) Ernst Haeckel (4) Ernst Mayr

134. Archaebacteria which degrade the cellulose in rumen (gut) of cattles is a/an :-

(1) Mycoplasma (2) Eubacterium

(3) Methanogen (4) Halophile

135. Which statement/s is/are not correct with respect to archaebacteria :-

(a) Archaebacteria differ from other bacteria in having a different cell wall structure

(b) Feature of cell wall is responsible for their survival in extreme condition

(c) Halophiles survive in high salt concentration.

(d) They are the oldest living fossils

(e) Peptidoglycan and muramic acid present in their cell wall

(1) a and e (2) b and e

(3) a, b and c (4) only e

136. Read the following statement (A–D):-

(A) Fatty acid are saturated or unsaturated

(B) Lecithin is a phospholipid which is present in plasma membrane

(C) Melting point of oils are high. Hence they remain liquid in winter

(D) The molecular weight of lipid do not exceed 800 dalton but they are present in acid insoluble pool.

How many statement are correct ?

(1) A,B,C,D (2) A,B,D

(3) B,C (4) A,C

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137. Consider the following four statements (a-d) and select the option which includes all the correct ones only :-

(a) Lipids are generally water insoluble.

(b) The R group could be a methyl ($-CH_3$) or ethyl ($-C_2H_5$) or higher no. of $-CH_2$ group (1 carbon to 19 carbon) in fatty acids

(c) Palmitic acid has 16 carbons including the carboxyl carbon.

(d) Arachidonic acid has 21 carbon atom including the carboxyl carbon.

Option

(1) Statement (a), (b) and (c)

(2) Statement (a), (b) only

(3) Statement (c), (d) only

(4) Statement (a), (c) and (d)

138. Given below is the structure of :-

$$CH_3-(CH_2)_{14}-\underset{\underset{O}{\|}}{C}-OH$$

(1) oleic acid

(2) Galactose

(3) Arachidonic acid

(4) Palmitic acid

139. Which of the following is not a macromolecule?

(1) Nucleic acid

(2) Polysaccharides

(3) Lipids

(4) Proteins

140. Lipids are extracted with acid insoluble fraction because

(1) Its molecular weight is more than 800 Da

(2) It form vesicle or micelle which are not water soluble

(3) It is a polymer

(4) It is a heteropolymer

141. Which of the following is homopolysaccharide?

(1) Cellulose

(2) Inulin

(3) Starch

(4) All of these

142. Which of the following secondary metabolites belong to toxin category?

(1) Codeine

(2) Concanavalin-A

(3) Anthocyanin

(4) Ricin

143. Which one protein shows incorrect match with its function :

(1) Collagen – Inter cellular ground substance.

(2) GLUT-4 - Enables glucose transport into cell

(3) Receptor – Sensory reception

(4) Trypsin – Hormone

144. The correct order of chemical composition of living tissues/cells in term of % of the total cellular mass is

(1) Nucleic acid > Proteins > H_2O > Carbohydrate > Ions > LIpid

(2) H_2O > Proteins > Nucleic acid > Carbohydrate > Lipid > Ion

(3) H_2O > Proteins > Carbohydrate > Nucleic acid > Lipid > Ions

(4) Lipid > Ions > Carbohydrate > H_2O > Proteins > Nucleic acid

145. Choose the element which is negligible in Human body:-

(1) Si (2) Mg

(3) Ca (4) O

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146. The two functional groups found in sugars are:-

(1) hydroxyl and methyl

(2) carbonyl and methyl

(3) carbonyl and phosphate

(4) carbonyl and hydroxyl

147. The compound having identifiable functions and play known roles in normal physiological processes are called :-

(1) Primary metabolites

(2) Secondary metabolites

(3) Biomacromolecules

(4) Polymeric substances

148. Read the following statements and find out the incorrect statement :-

(1) Secondary metabolites are found in plant, fungal and microbial cells.

(2) Many secondary metabolites like rubbers, drugs, spices, scents and pigments are useful to human welfare.

(3) Some secondary metabolites have ecological significance.

(4) Flavonoids, antibiotics, coloured pigments and lecithin are secondary metabolites.

149. Which of the following diasaccharide gives two molecules of glucose on hydrolysis ?

(1) Maltose

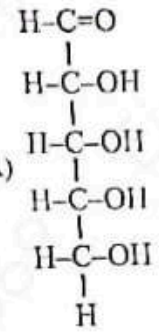
(2) Lactose

(3) (1) and (2) both

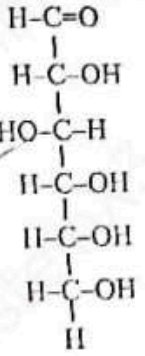
(4) Sucrose

150. Which one out of A-D given below correctly represents the structural formula of the glucose :-

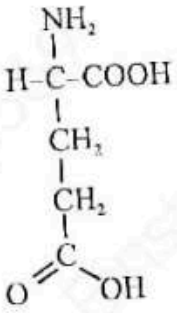
(A)



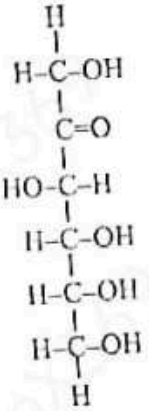
(B)



(C)



(D)



(1) B (2) A (3) C (4) D

151. Which is a derived lipid?

(1) linoleic acid (2) Cholesterol

(3) Glycerol (4) Lecithin

152. Match the column-I with column-II :-

	Column-I		Column-II
A	Hexoses	I	Starch
B	Disaccharides	II	Fructose
C	Polysaccharides	III	Lactose
D	Mucopolysaccharides	IV	Heparin

Option :-

	A	B	C	D
(1)	I	II	III	IV
(2)	II	III	I	IV
(3)	II	III	IV	I
(4)	II	I	III	IV

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153. Which of the following statements are not true?

(A) Glycerol is a 3-carbon compound with three–OH groups.

(B) Waxes are monoesters with only one molecule of fatty acid attached along with chain of monohydroxy alcohol.

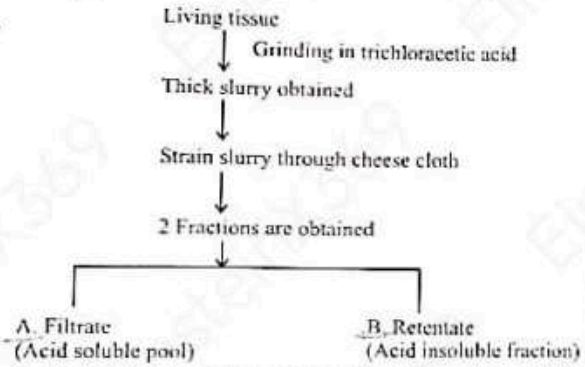
(C) Proteins are heteropolymers of nucleotides.

(D) Stearic acid is a unsaturated fatty acid while oleic acid is saturated fatty acid.

(1) A & B only (2) B & C only

(3) C & D only (4) A & D only

154.



I. Molecular weight ranging from less than 1000 Dalton approximately

II. Has four type of organic compounds proteins, nucleic acids, polysaccharides and lipids

III. Contain chemicals that have molecular weight more than 1000 Dalton

IV. Has monomers

V. Has generally polymers except lipid

VI. Mostly micromolecules

VII. Generally macromolecules

How many of the above statements (I-VII) are correct regarding A & B respectively ?

	A	B
(1)	I, II, III	IV, V, VI, VIII
(2)	II, IV, VI	I, III, V, VII
(3)	I, IV, VI	II, III, V, VII
(4)	I, III, V	II, IV, VI, VII

155. N-acetyl galactosamine is ?

(1) Modified polysaccharide

(2) Modified sugar

(3) Derivative of protein

(4) Amino acid

156. Which of the following is an essential amino acid ?

(1) Cysteine

(2) Tyrosine

(3) Tryptophan

(4) Glycine

157. All of the following are made up of cellulose except :

(1) Plant cell wall

(2) Cotton fibres

(3) Exoskeleton of arthropods

(4) Paper made from plant pulp

158. Read the following statements (A – E) and answer as asked next to them :-

(A) Peptide bonds are not broken in denaturation of protein

(B) Protein denaturation is a type of biophysiochemical change

(C) Both +ve and –ve charge are present on amino acid in equal amount at isoelectric point

(D) Haemoglobin is a type of protein

(E) Tertiary configuration of protein is soluble in water.

How many of above statements are correct?

(1) 3

(2) 4

(3) 5

(4) 1

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159. Read the following statements. Find out how many statements are correct.

(A) RuBisCO is the most abundant protein in the whole of the biosphere

(B) The acid soluble pool represents roughly the cytoplasmic composition.

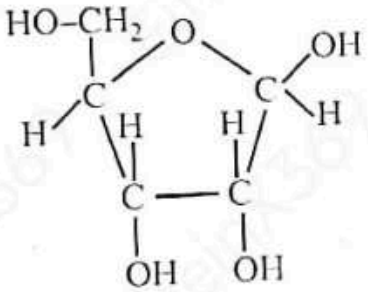
(C) The neural tissues have lipids with more complex structures.

(D) Biomicromolecules are polymers

(1) Four (2) Three

(3) Two (4) One

160. Following structure is related to which biomolecule?



(1) α-Glucose (2) β-Glucose

(3) α-Ribose (4) β-Ribose

161. Which statement is incorrect?

(1) Diterpene is secondary metabolite.

(2) Proteins are 10-15% in a cell.

(3) Insulin is an enzyme.

(4) Some protein are hormones.

162. How many nitrogen atoms are present in Lecithin?

(1) 4 (2) 3 (3) 2 (4) 1

163. Secondary metabolites are observed in

(1) plant cells

(2) fungal cells

(3) microbial cells

(4) all of these

164. Carbohydrates are -

(1) Polymers of fatty acids

(2) Polymers of amino acids

(3) Polyhydroxy aldehydes or ketones

(4) None

165. Which of the following is the simplest amino acid –

(1) Alanine (2) Asparagine

(3) Glycine (4) Tyrosine

166. Which of the following is heteropolymer ?

(1) Protein

(2) Pectin

(3) Cellulose

(4) Both (1) and (2)

167. Given below are two statements:

Statement I: The acid insoluble pool represents roughly the cytoplasmic composition.

Statement II: Proteins are linear chains of amino acids linked by peptide bonds.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I and Statement II both are correct.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Statement I and Statement II both are incorrect.

168. Select the element that is found in higher amount in human body than earth's crust.

(1) Sodium (2) Sodium

(3) Magnesium (4) Carbon

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169. Choose the odd one w.r.t. aromatic amino acids

(1) Glycine (2) Tyrosine

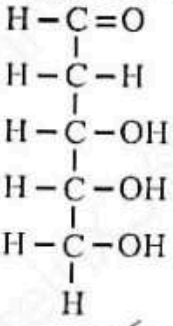
(3) Tryptophan (4) Phenylalanine

170. Find the wrong match pair :-

(1)	Secondary metabolites	Anthocyanin
(2)	Chitin	Polysaccharide
(3)	Cellulose	Heteropolymer
(4)	Protein	Insulin

(1) 1 (2) 2 (3) 3 (4) 4

171. Following structure is related to which compound ?



(1) Ribose (2) Deoxyribose

(3) Glucose (4) Fructose

172. When a long protein chain is folded upon itself like a hollow woollen ball, it gives rise to

(1) Primary structure of protein

(2) Secondary structure of protein

(3) Tertiary structure of protein

(4) Quaternary structure of protein

173. Read the following statements and find out the incorrect statement.

(1) Oil have lower melting point (e.g. gingely oil) and hence remain as oil in winters

(2) Glycosidic bond found only polysaccharide

(3) Triglycerides are formed by esterification of Glycerol with three fatty acids

(4) Both (1) & (3)

174. Match the following with correct combination

	Column - I		Column - II
A.	Triglycerides	1	Galactose
B.	Lactose	2	Glycerol
C.	RNA	3	Palmitic acid
D.	β pleated	4	Uracil
E.	Beeswax	5	Secondary structure

(1) A-4, B-1, C-5, D-2, E-3

(2) A-5, B-1, C-4, D-2, E-3

(3) A-3, B-1, C-4, D-5, E-2

(4) A-2, B-1, C-4, D-5, E-3

175. Unbranched polymer of glucose is

(1) Starch (2) Glycogen

(3) Cellulose (4) Inulin

176. A protein that has three helical polypeptides supercoiled to form a rope like structure is

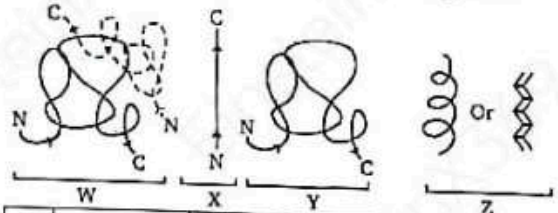
(1) Insulin

(2) Collagen

(3) Haemoglobin

(4) (1) and (3)

177. Find out the primary, secondary, tertiary and quaternary protein structures in the diagram :

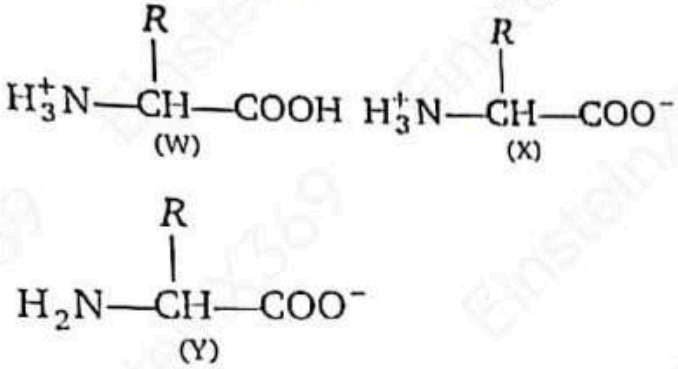


	W	X	Y	Z
(1)	Quaternary	Secondary	Primary	Tertiary
(2)	Tertiary	Primary	Quaternary	Secondary
(3)	Quaternary	Primary	Tertiary	Secondary
(4)	Secondary	Primary	Quaternary	Tertiary

(1) 1 (2) 2 (3) 3 (4) 4

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178. Which of the following is a zwitter ion ?



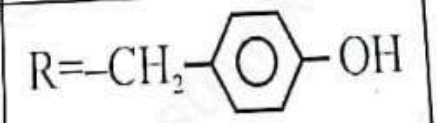
(1) W

(2) W and X

(3) W and Y

(4) X

179. Match the column P to Q

	Column-P		Column-Q
A	Glutamic acid	i	R=$-CH_2-CH_2-COOH$
B	Cysteine	ii	R=$-CH_2-SH$
C	Tyrosine	iii	R=$-CH_2-OH$
D	Serine	iv	

(1) A-i, B-ii, C-iv, D-iii

(2) A-i, B-ii, C-iii, D-iv

(3) A-iv, B-iii, C-ii, D-i

(4) A-ii, B-i, C-iii, D-iv

180. Arachidonic acid is :-

(1) Unsaturated fatty acid

(2) Essential amino acid

(3) Essential fatty acid

(4) Both (1) & (3)

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